

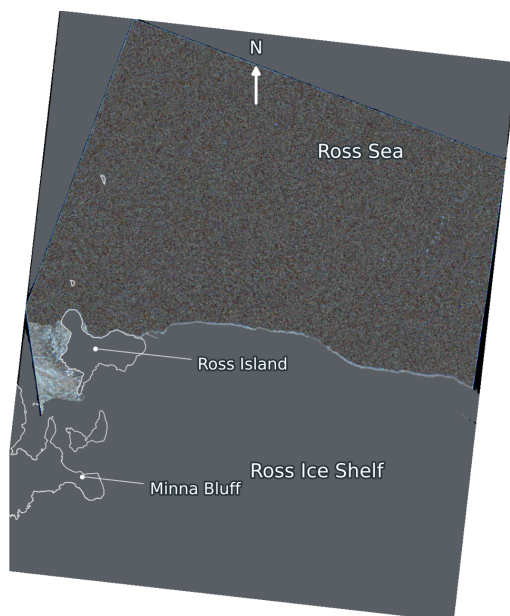
Antarctic Fast Ice Watch

Primary and additional observation products

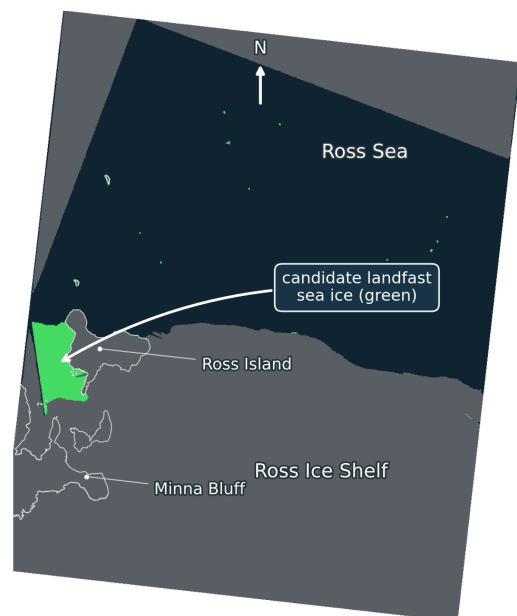
A weekly one-page bulletin for the Casey, Davis and Mawson approaches. The primary product will be Sentinel-1 fast-ice extent and change maps, updated with suitable overlapping acquisitions (target approximately every four days, subject to coverage and quality). The additional product will be SWOT freeboard maps; thickness estimation will be investigated through validation and explicit uncertainty assessment.

PRIMARY PRODUCT | FAST-ICE EXTENT AND CHANGE MAPS

a) composite SAR



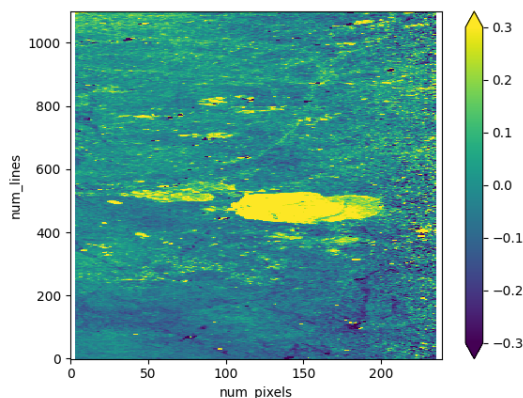
b) classification



10–22 October 2021 | North aligned at centre (173.52°E) | Grey: masked land / out-of-swath areas
Faint lines: grounded-land outlines — SCAR Antarctic Digital Database, 2026 (contextual reference)

Research workflow example: SAR texture composite and predicted classification, 10–22 October 2021, McMurdo/Ross Sea. Green: candidate landfast sea ice (class 2; originally blue). Grey: masked land/ice shelf and out-of-swath areas. Faint lines: grounded-land outlines from SCAR Antarctic Digital Database, 2026, provided as geographic context. Both panels share orientation and crop. This extent example does not show change.

ADDITIONAL PRODUCT | SWOT FREEBOARD MAPS



Research example: Alex Fraser's Ross Sea demonstration, SWOT cycle 056 / pass 142, acquired 15 September 2026. The notebook plots sea-surface-height anomaly (`ssha_kar in_2`, metres), with a quadratic cross-track profile removed. Axes are pixel/line indices; range -0.3 to +0.3 m. This demonstrates relative elevation, not validated freeboard or thickness. The planned additional product requires rigorous retrieval, quality control and uncertainty assessment.